Calculus II

Tutorial #8 07–11 Apr 2025

Topics:

- Initial-value problems
- Separable differential equations

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. The general solution to a separable differential equation contains one constant whose value can be determined from an initial condition for the differential equation.
2. The differential equation
$$\frac{dy}{dx} = x\sin(xy)$$
is separable.
3. The differential equation
$$\frac{dy}{dx} = e^{x+y}$$
is separable.

• **Problems**.

4. Solve the following IVP:
   (a) $y' = e^{2x} + \sqrt{x} + \cos(3x),\ y(0) = 2$
   (b) $y'' = xe^x,\ y(0) = 3,\ y'(0) = 4$
5. Determine the existence and uniqueness of solution to the initial-value problem
$$\frac{dy}{dx} = 3xy^{1/3}, \quad y(0) = a.$$
where a is a real non-zero parameter.
6. Even simple-looking differential equations can have complicated solution curves. Let's study solution curves of the differential equation
$$y' = -2xy^2. \tag{5}$$

(a) Verify that the hypotheses of the existence and uniqueness theorem are satisfied for the IVP

$$y' = -2xy^2, \quad y(x_0) = y_0$$

for every (x_0, y_0). This shows that the IVP always has a unique solution on some interval containing x_0.

(b) Verify that for all values of the constant C, $y(x) = \dfrac{1}{x^2 + C}$ is a solution to (5).

(c) Use the solution in (b) to solve the following IVPs. For each case, determine the maximum interval on which your solution is valid.

(i) $y' = -2xy^2, \quad y(0) = 1.$
(ii) $y' = -2xy^2, \quad y(1) = 1.$
(iii) $y' = -2xy^2, \quad y(0) = -1.$

(d) What is the unique solution to the following IVP?

$$y' = -2xy^2, \quad y(0) = 0.$$

7. Solve the following differential equations.

(a) $ydx - (x-2)dy = 0$

(b) $\dfrac{dy}{dx} = \dfrac{x(y^2-1)}{2(x-2)(x-1)}$

8. Solve the differential equation

$$\sec x \, \frac{dy}{dx} = e^{y+\sin x}$$

9. Solve the differential equation

$$y^2 \, \frac{dy}{dx} = 3x^2y^3 - 6x^2$$

10. Solve the differential equation

$$\frac{dy}{dx} = \sqrt{y}\cos^2\sqrt{y}$$

END

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Answers to selected problems

1. True 2. False 3. True
4. (a) $\dfrac{1}{2}e^{2x} + \dfrac{2}{3}x^{3/2} + \dfrac{1}{3}\sin(3x) + \dfrac{3}{2}$

6. (c) [i] $x \in \mathbb{R}$ [ii] $x \in (0, \infty)$ [iii] $x \in (-1, 1)$

7. (a) $y = C(x-2)$ (b) $\dfrac{(x-1)+C(x-2)^2}{(x-1)-C(x-2)^2}$ and $y = -1$

8. $-\ln\left(C - e^{\sin x}\right)$

9. $\sqrt[3]{2 + Ce^{3x^3}}$

10. $y = 0, \quad y = \left(n\pi + \dfrac{\pi}{2}\right)^2 \quad (n = 0, 1, 2, 3, \ldots), \quad 2\tan(\sqrt{y}) = x + C.$